# SMARTX: Porewater Ammonium

May 2025 Samples

2025-10-22

## Contents


0.1 Import Data & Clean . . . 3
0.2 Assessing standard Curves . . . 3
0.3 Dilution Corrections - ensure the latest dilution is kept . . . 5
0.4 Performance Check . . . 5
0.5 Analyze the Check Standards . . . 6
0.6 Analyze Blanks . . . 7
0.7 Analyze Duplicates . . . 8
0.8 Spikes . . . 9
0.9 Matrix Effects . . . 10
0.10 Unit Converted Data Column Added (mg/L to uM ) . . . 10
0.11 Sample Flagging - Within range of standard curve . . . 10
0.12 Merge samples with Metadata & check all samples are present . . . 10
0.13 Visualize Data . . . 11
0.14 Export Processed Data . . . 15

##Run Information

```r
cat("Run Information:  ") #lets you know what section you're in
```

```
## Run Information:
```

```r
#set the run date & user name
  run_date <- "08/12/2025"
  sample_year <- "2025"
  sample_month <- "May"
  user <- "Victoria Ellis"

#identify the files you want to read in
  #read in as a list to accommodate multiple runs in a month
  NH3_files <- c("Raw Data/SMARTX_May_2025_Run1.csv",
                 "Raw Data/SMARTX_May_2025_Run2.csv",
                 "Raw Data/SMARTX_May_2025_Run3.csv")

# Define the file path for QAQC log file - NO Need to change just check year
  file_path <- "Raw Data/SEAL_SMARTX_QAQC_Log_2025.csv"
  final_path <- "Processed Data/GCReW_SMARTX_Porewater_NH4_202505.csv"

#record any notes about the run or anything other info here:
  run_notes <- "PE Check contaminated, NH3 high, PE Check changed after this run
                Moved metadata chunk to setup because 'here' package was needed"

  #Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

  cat(run_notes)
```

```
## PE Check contaminated, NH3 high, PE Check changed after this run
##                 Moved metadata chunk to setup because 'here' package was needed
```

##Setup

##Read in metadata and create similar sample IDs for matching to samples

## 0.1 Import Data & Clean

## 0.2 Assessing standard Curves

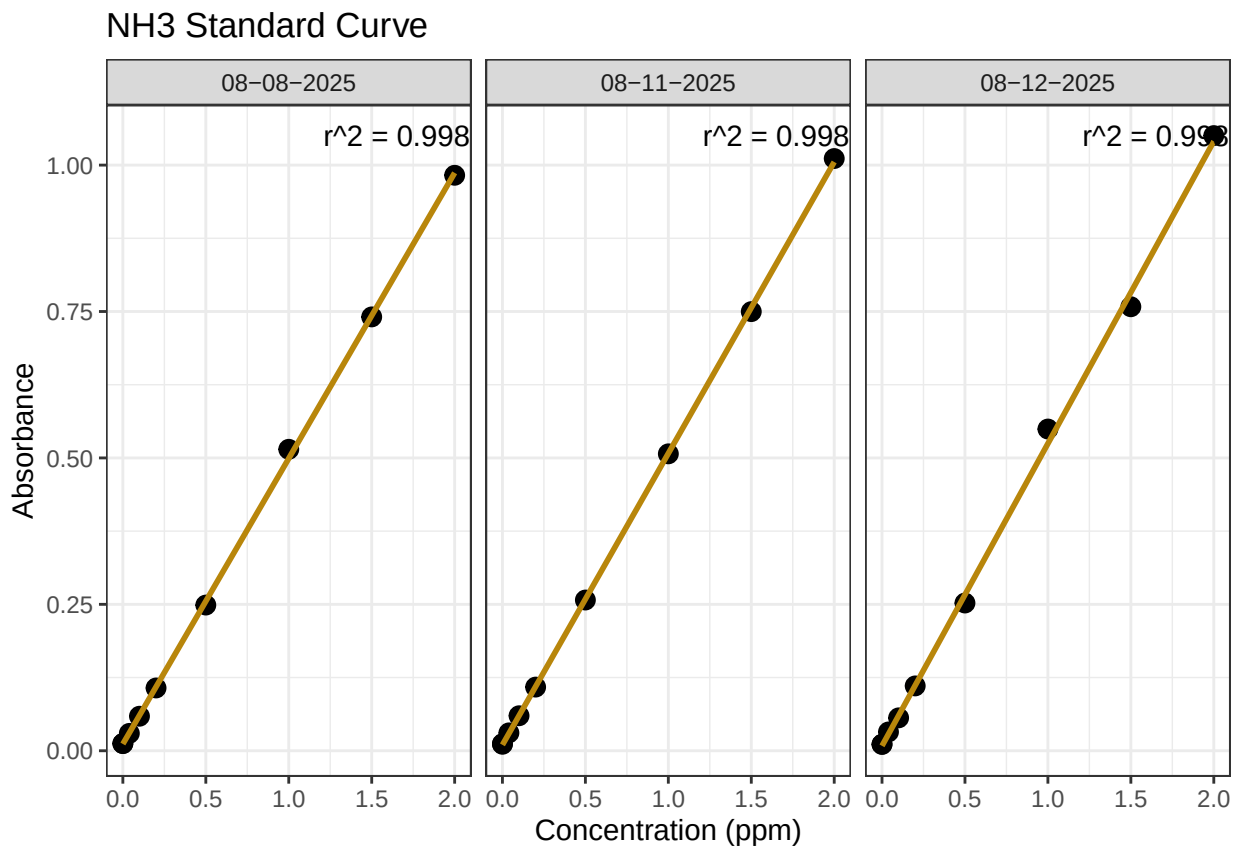
#Pull out standards data

```
## Assess Standard Curves
```

#Plot standards data

```
## Assess Standard Curves
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## [1] "NH3 Curve r2 GOOD - PROCEED"
```

```
## [1] "QAQC log file exists and has been read into the code."
```

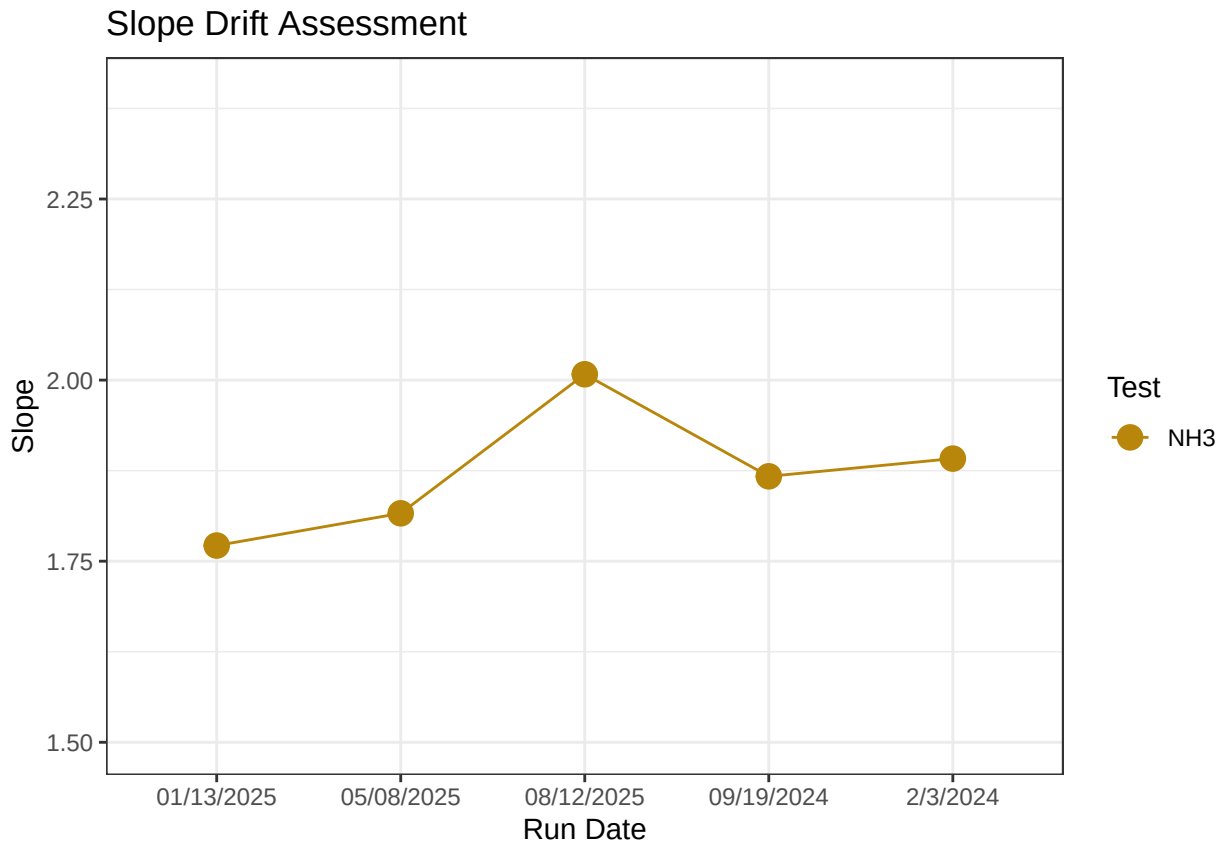



Table 1: Average Slope by Analyte

| Test | avg_slope |
|---|---|
| NH3 | 1.871 |

## 0.3 Dilution Corrections - ensure the latest dilution is kept

```
## Dilution Corrections

## Duplicated samples: , SMARTX_442-40, SMARTX_442-80, SMARTX_443-40, SMARTX_443-80, SMARTX_443-120, SMA

## Duplicated samples with NO dilution present (possible input error): , 10ppt, pe_chk
```

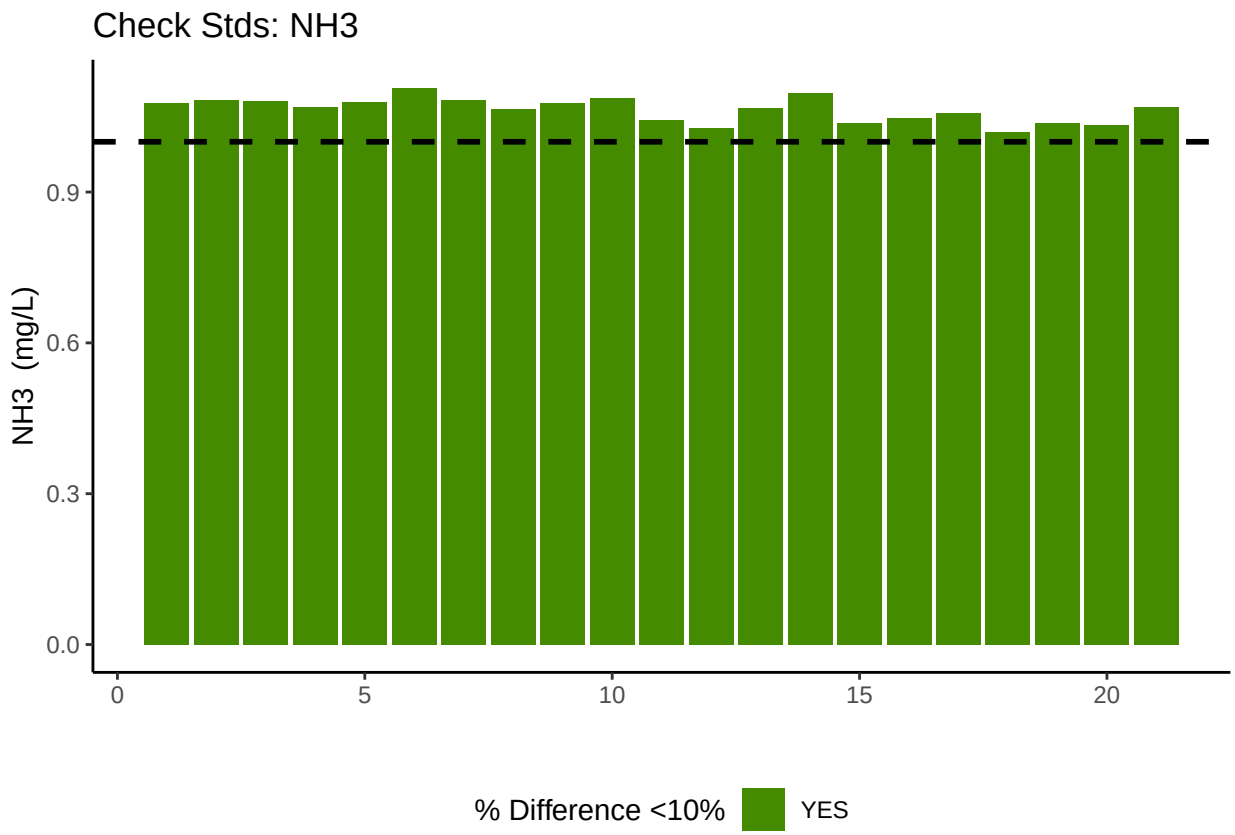
## 0.4 Performance Check

```
## [1] "NH3 pe Check has a % Difference >25% - REASSESS"

## Run mean = 1.40697

## Expected  = 1.034
```

## 0.5 Analyze the Check Standards

```
## Analyze Check Standards
```

```
## [1] "NH3 Check Standard RSD within Range - PROCEED"
```




```
## [1] ">60% of NH3 Check Standards are within range of expected concentration - PROCEED"
```

## 0.6 Analyze Blanks

```
## Assess Blanks
```

```
## [1] ">60% of NH3 Blanks are below the lower 25% quartile of samples - PROCEED"
```

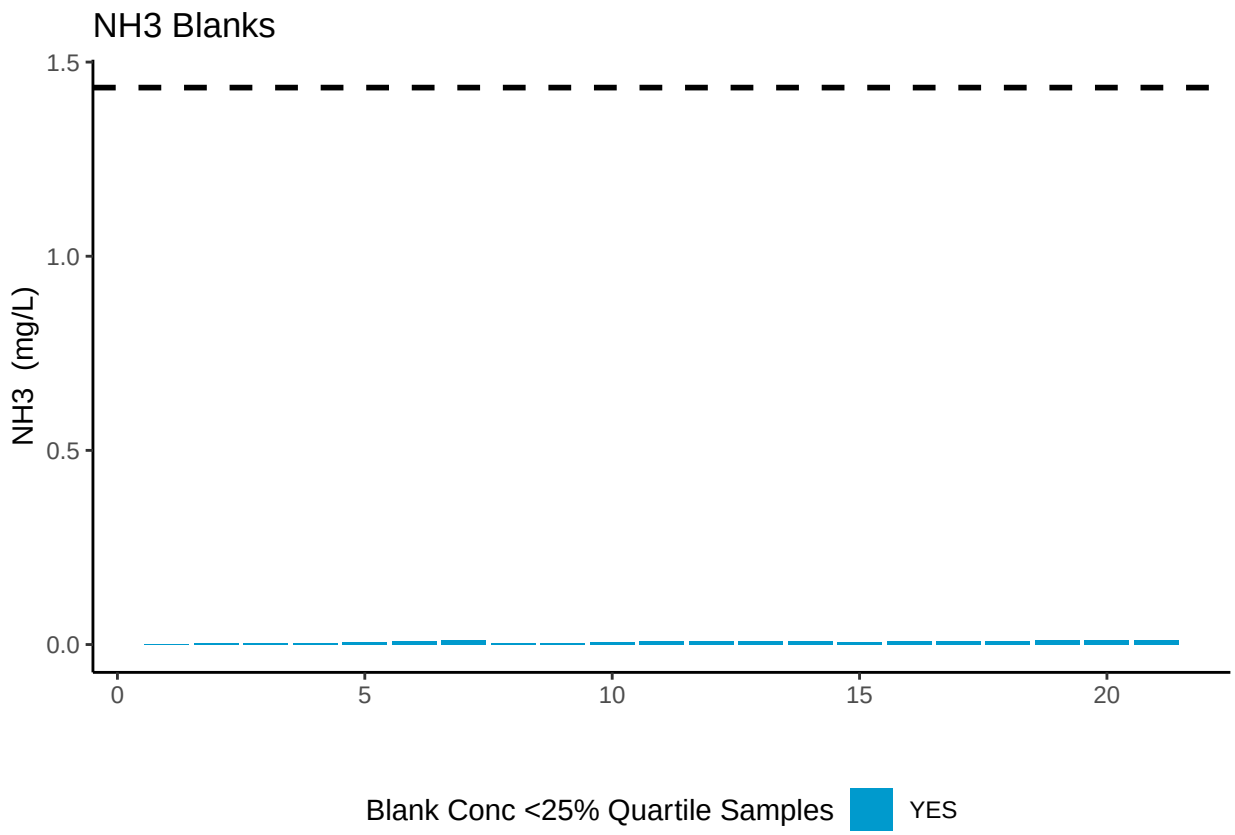



Table 2: Mean Concentration of Blanks

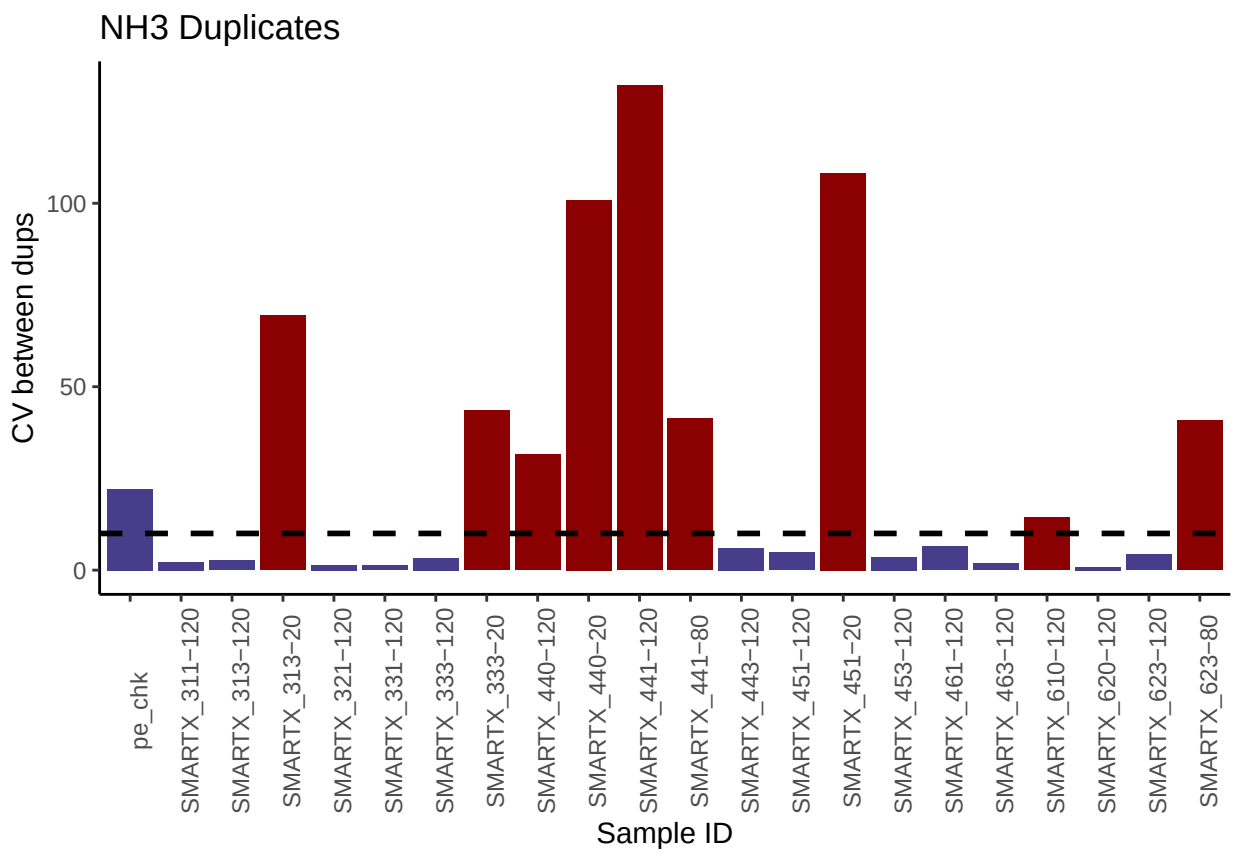
| Test | Blank_Mean_Conc |
|---|---|
| NH3 | 0.0074 |

## 0.7 Analyze Duplicates

```
## Analyze Duplicates
```

```
## Warning: Returning more (or less) than 1 row per ‘summarise()‘ group was deprecated in
## dplyr 1.1.0.
## i Please use ‘reframe()‘ instead.
## i When switching from ‘summarise()‘ to ‘reframe()‘, remember that ‘reframe()‘
##   always returns an ungrouped data frame and adjust accordingly.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

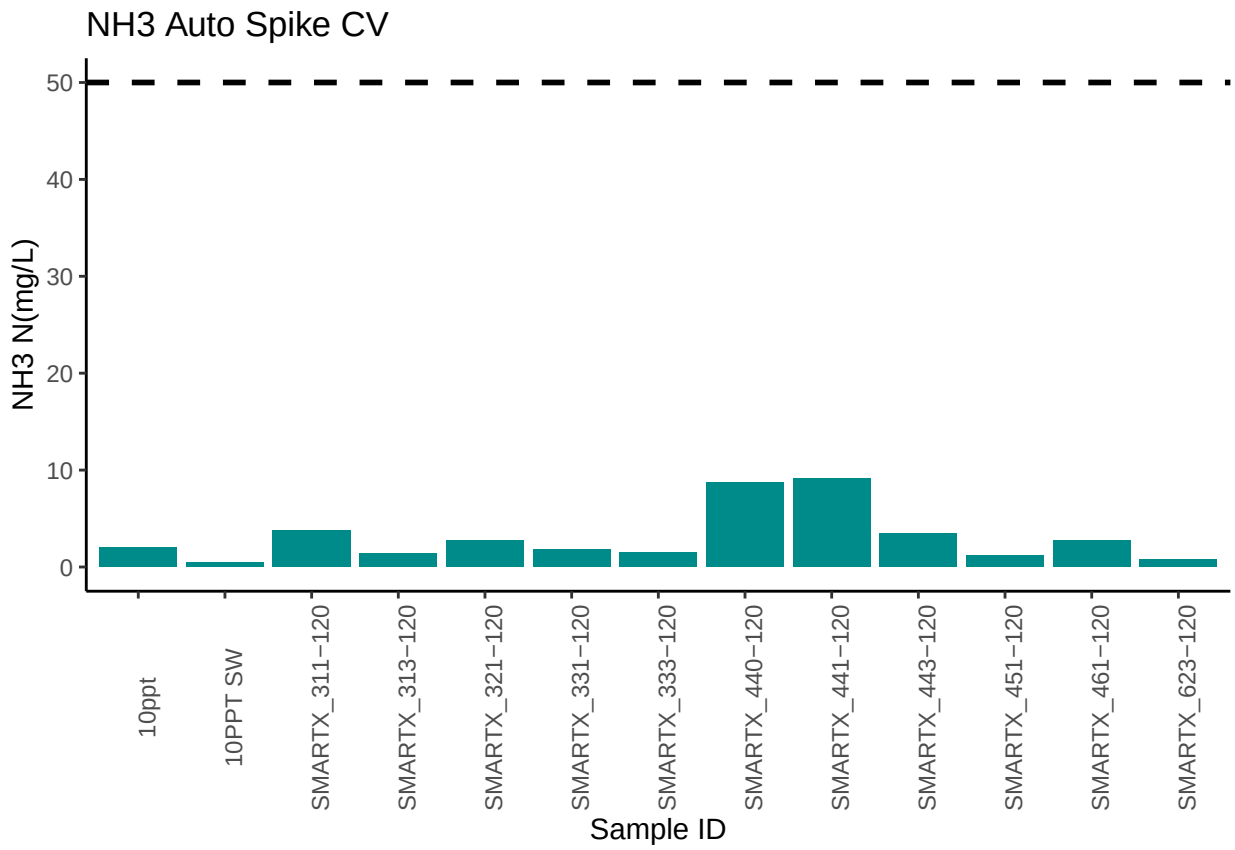
```
## [1] ">60% of NH3 Duplicates have a CV <10% - PROCEED"
```



## 0.8 Spikes

```
## [1] ">60% of Spikes have a CV <50% - PROCEED"
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



## 0.9 Matrix Effects

```
## [1] "NO NH3 Matrix Effect, PROCEED"
```

## 0.10 Unit Converted Data Column Added (mg/L to uM )

## 0.11 Sample Flagging - Within range of standard curve

```
## Sample Flagging
```

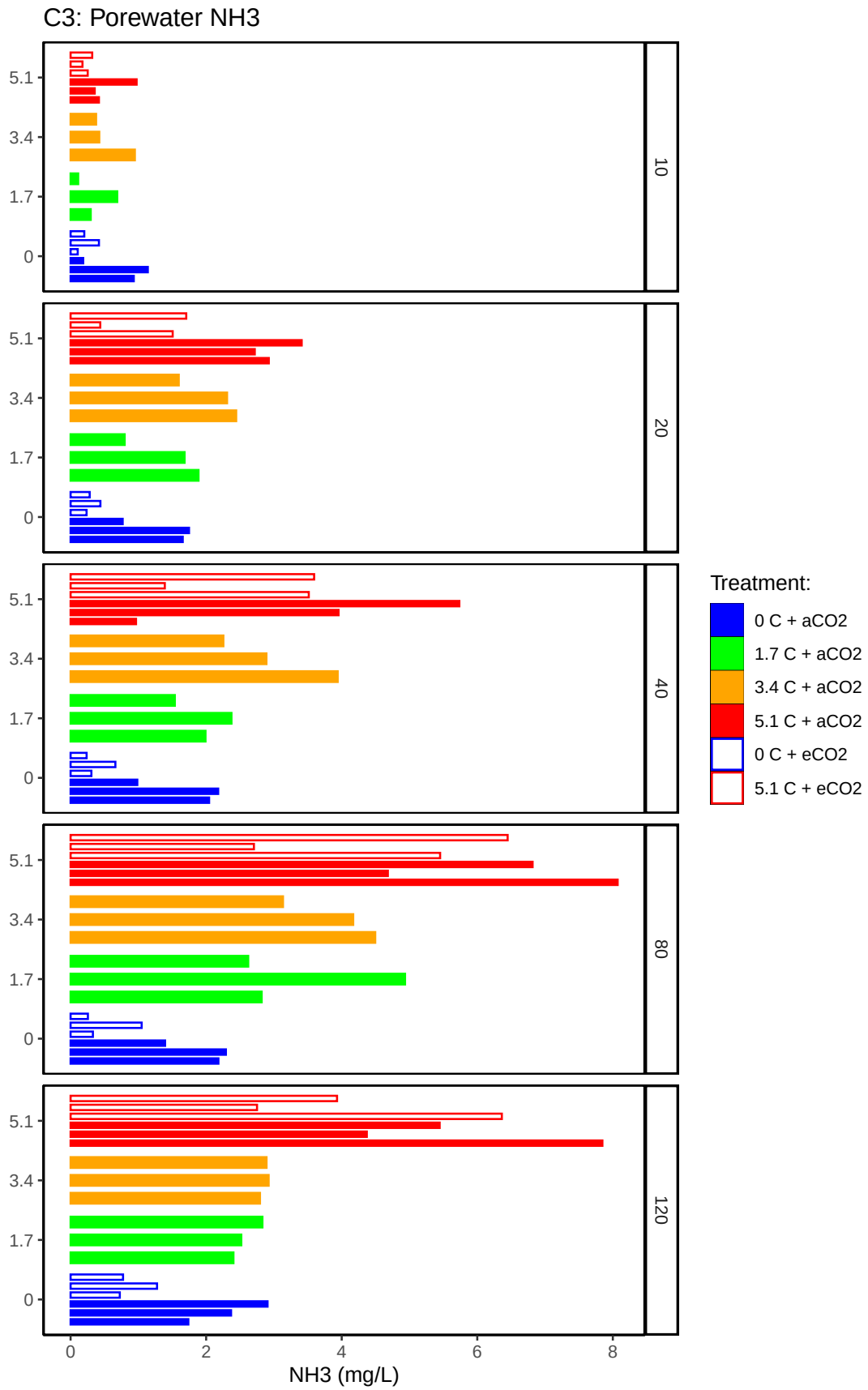
## 0.12 Merge samples with Metadata & check all samples are present

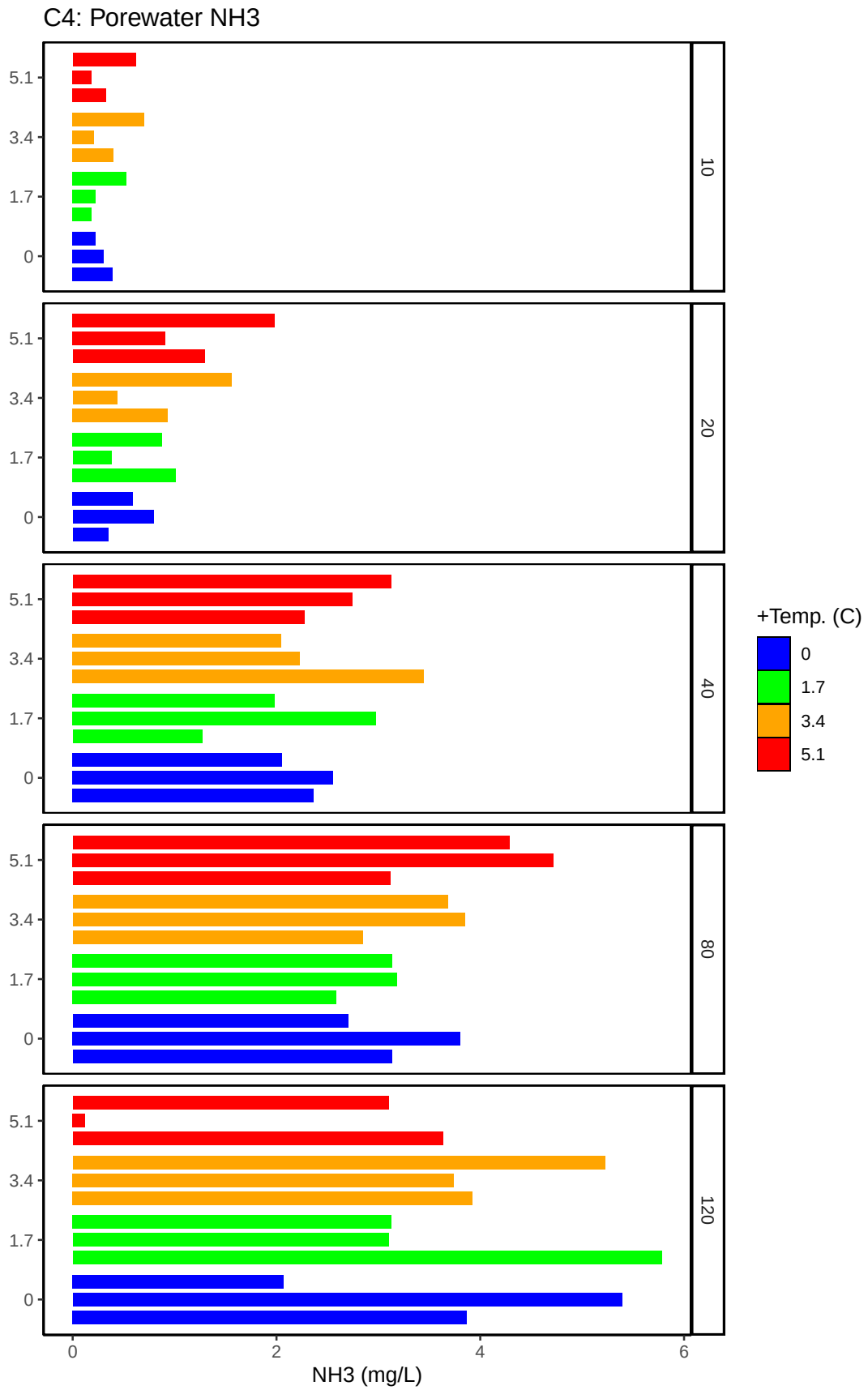
```
## Merge Samples with Metadata
```

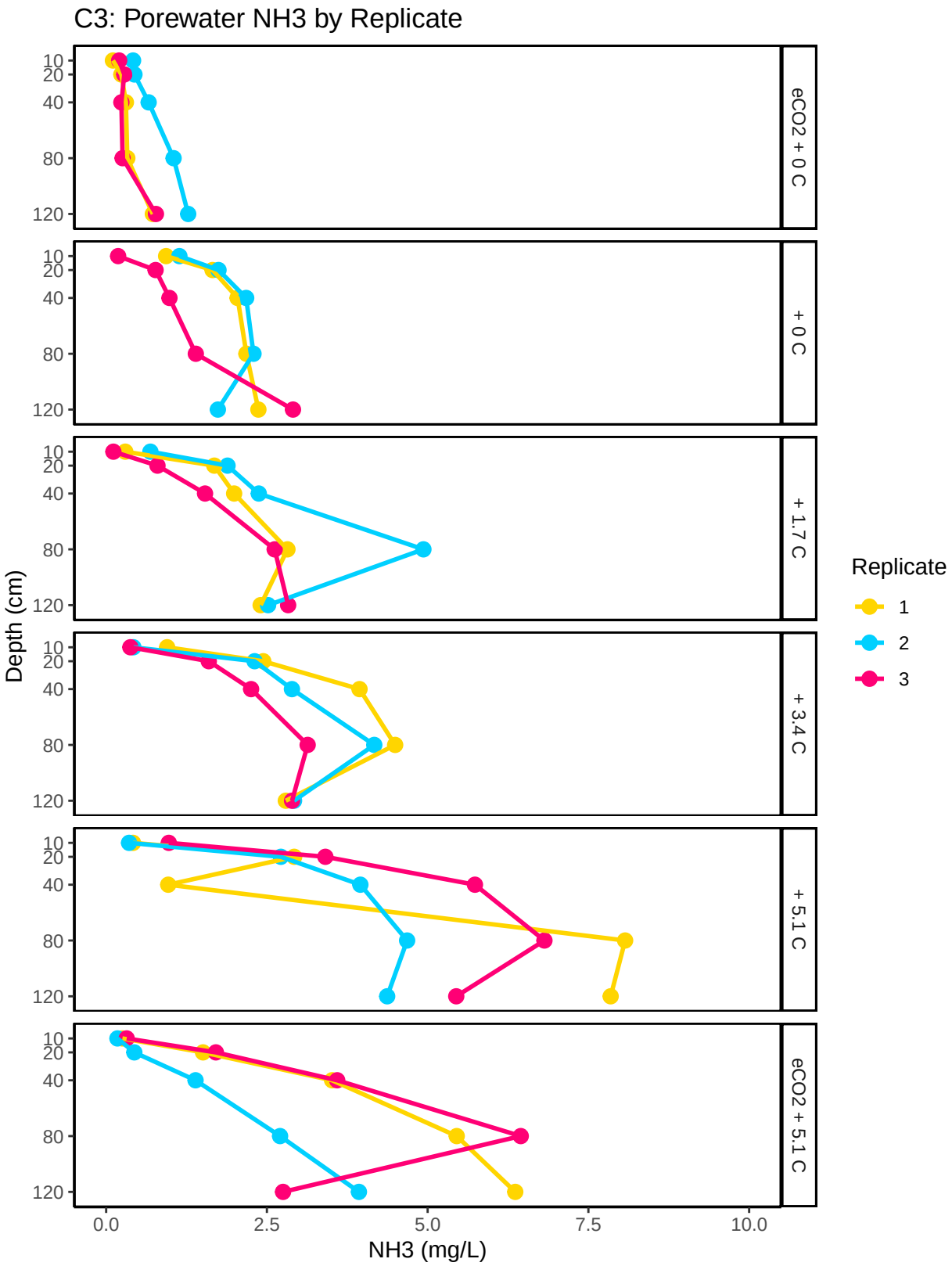
```
## All sample IDs are present in data.
```

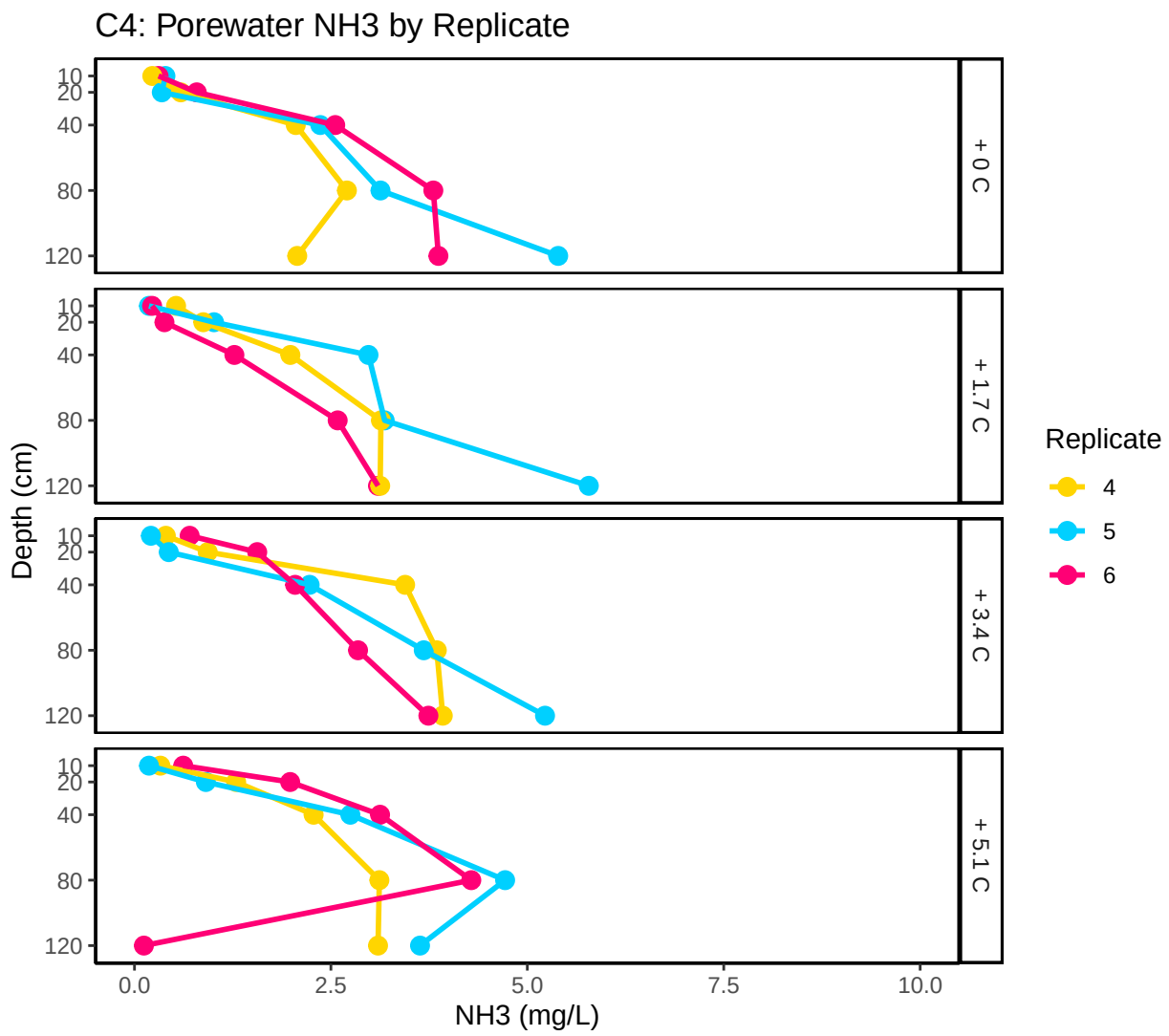
## 0.13 Visualize Data

```
## Visualize Data
```


C3: Porewater NH3
5.1
3.4
1.7
0
10
20
40
80
120
0
2
4
6
8
NH3 (mg/L)
Treatment:
0 C + aCO2
1.7 C + aCO2
3.4 C + aCO2
5.1 C + aCO2
0 C + eCO2
5.1 C + eCO2


C4: Porewater NH3
5.1
3.4
1.7
0
10
20
40
80
120
0
2
4
6
NH3 (mg/L)
+Temp. (C)
0
1.7
3.4
5.1


C3: Porewater NH3 by Replicate
Depth (cm)
10
20
40
80
120
eCO2 + 0 C
+ 0 C
+ 1.7 C
+ 3.4 C
+ 5.1 C
eCO2 + 5.1 C
0.0
2.5
5.0
7.5
10.0
NH3 (mg/L)
Replicate
1
2
3




## 0.14 Export Processed Data

#end